

Vector-Space Optimization Framework Based on Projected Contraction Condition for Control Design with Input Saturation

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1 Motivation

- Introduction to Contraction Theory
- Existing Approaches
- Common Limitations
- Summary and Motivation for Proposed Method

2 Problem Formulation and Preliminaries

- Problem Formulation
- Revisit of Existing Approach (CV-STEM)

3 Proposed Method

- Vector-Space Optimization Framework

4 Simulation Validation

- Simulation Setup
- Simulation Results

5 Conclusion

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Conclusion

Contraction Theory is a powerful tool for analyzing the stability of nonlinear systems by examining **differential dynamics**, providing insights into the behavior of **trajectories** (not points) and their convergence properties ^[1].

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) \quad \rightarrow \quad \text{diff. dyn. } \frac{d}{dt}\delta\mathbf{x} = \frac{\partial\mathbf{f}}{\partial\mathbf{x}}\delta\mathbf{x}. \quad (1)$$

[1] W. Lohmiller and J.-J. E. Slotine, "On contraction analysis for non-linear systems," *Automatica*, 1998

[2] H. Tsukamoto, S.-J. Chung, and J.-J. E. Slotine, "Contraction theory for nonlinear stability analysis and learning-based control: A tutorial overview," *Annual Reviews in Control*, 2021

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Typically used Lyapunov function is defined in terms of $\delta\mathbf{x}$ rather than $\mathbf{x}$, as follows:

$$V = \mathbf{x}^\top \mathbf{P}\mathbf{x} \rightarrow V = \delta\mathbf{x}^\top \mathbf{M}(\mathbf{x}, t)\delta\mathbf{x}. \quad (2)$$

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Introduction to Contraction Theory

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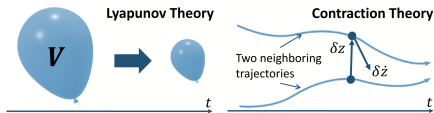


Figure: Lyapunov theory versus contraction theory ^[2].

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- **Stability Analysis:** The stability analysis can be simplified.

Therefore, in control design, **Contraction Theory** provides:

- **Exponential Convergence:** Trajectories globally exponentially converge *regardless of initial conditions*.
- **Control Design:** Owing to nature of differential dynamics, *LPV control design* methods can be applied.
- **Stability Analysis:** The stability analysis can be simplified.

Now, the question is how to use the contraction theory in control design?

1. **Stability Analysis Tools:** Stability under a given controller can be analyzed using contraction theory.
2. **Control Design Tools** (what we focus on): contraction theory can be used to design stabilizing controllers, e.g., CCM, CV-STEM, etc.

Existing Approaches Control Contraction Metric (CCM)

Control Contraction Metric (CCM) [1]

Consider a control-affine system and Lyapunov function candidate as follows:

$$\frac{d}{dt} \mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t) \mathbf{u} \rightarrow \text{diff. dyn. } \frac{d}{dt} \delta \mathbf{x} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \delta \mathbf{x} + \frac{\partial \mathbf{B}}{\partial \mathbf{x}} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u}, \text{ and } V = \delta \mathbf{x}^\top \mathbf{M}(\mathbf{x}, t) \delta \mathbf{x}, \quad (3)$$

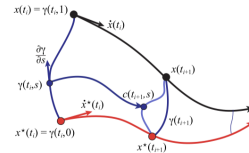
Theorem (Control Contraction Metric (CCM), informal)

For a given system, if there exists a bounded metric $\mathbf{M}(\mathbf{x}, t)$ such that holds

$$\frac{d}{dt} \mathbf{M} + 2 \text{sym} \left(\mathbf{M} \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right) \leq -2\alpha \mathbf{M}, \quad \left(\frac{d}{dt} V \leq -2\alpha V \right), \quad (4)$$

for all $\delta \mathbf{x} \neq 0$ such that $\delta \mathbf{x}^\top \mathbf{M} \mathbf{B} = 0$, then the system is **universally exponentially stabilizable** via state feedback [2].

By using $\mathbf{M}$, a feedback controller $\mathbf{u}$ can be established considering the closest path between the current and target states in the **Riemannian space** defined by $\mathbf{M}$, i.e., path integral of $\delta \mathbf{u} = k(\mathbf{x}, \delta \mathbf{x}, \mathbf{u}, t)$.



[1] I. R. Manchester and J.-J. E. Slotine, "Control contraction metrics: Convex and intrinsic criteria for nonlinear feedback design," IEEE TAC, 2017

[2] H. Tsukamoto, et al., "Contraction theory for nonlinear stability analysis and learning-based control: A tutorial overview," Annual Reviews in Control, 2021

CMM can provide theoretical guarantees for the existence of stabilizing controllers, which can reshape the system dynamics to be contracting.

However, ...

- Finding contraction metric M is **computationally expensive** (should be solved using SOS, LMIs, etc. using sampled data points in interested region).
- No consideration of input saturation and not straightforward to incorporate it.

ConVex optimization-based Steady-state Tracking Error Minimization (CV-STEM)^[1].

According to Theorem 2.4 in [1], by **minimizing the condition number** χ of $\mathbf{M}$, the **steady state error** can be reduced.

$$\downarrow \chi(\text{condition number of } \mathbf{M}) \rightarrow \downarrow \text{steady state error} \quad (5)$$

The control input can be simply obtained by

$$\mathbf{u} = \mathbf{u}_d - \mathbf{R}^{-1} \mathbf{B}^\top \mathbf{M}(\mathbf{x} - \mathbf{x}_d), \quad (\mathbf{M} \text{ is obtained by } \text{solving LMIs which minimizes } \chi.) \quad (6)$$

Main features of CV-STEM are as follows:

- Easy to implement (step-wise convex condition and affine control law).
- Due to the use of LMIs, it should be solved in advance.
- Still has no consideration of input saturation.

^[1]H. Tsukamoto, et al. , "Neural stochastic contraction metrics for learning-based control and estimation," IEEE Control Systems Letters, 2021

What if the control input is saturated?

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t)\text{sat}(\mathbf{u}) \quad (7)$$

Typically, contraction theory-based control design methods does not consider input constraints, which can lead to **undesirable behavior** in practical applications.

Furthermore, the existing methods compute $\mathbf{M}$ using matrix-space optimization problems, explicitly imposing constraints in input vector space is not straightforward.

The proposed approach ...

- Transforms the **matrix-space optimization problem** for M into a **vector-space optimization problem** by **projecting the contraction condition** into a vector space.
- Handles the input saturation in a more direct way by **incorporating the input saturation** into the optimization problem.

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Consider a control-affine system under input saturation and external disturbance:

$$\frac{d}{dt} \mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t) \text{sat } \mathbf{u} + \mathbf{d}(t), \quad (8)$$

assuming that

- The system functions are smooth implying Lipschitz continuity.
- The control effectiveness matrix $\mathbf{B}(\mathbf{x}, t)$ is full column rank.
- The disturbance is bounded as $\|\mathbf{d}(t)\| \leq \bar{d}$.

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The desired trajectory is supposed to be given as $\mathbf{x}_d(t)$ and $\mathbf{u}_d(t)$, which satisfies the system dynamics:

$$\frac{d}{dt} \mathbf{x}_d = \mathbf{f}(\mathbf{x}_d, t) + \mathbf{B}(\mathbf{x}_d, t) \mathbf{u}_d, \quad (9)$$

without input saturation and external disturbance.

Problem Formulation

Consider a control-affine system under input saturation and external disturbance:

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Problem Statement

The objective is to design a stabilizing controller $\mathbf{u}$ such that the error dynamics $\mathbf{e} = \mathbf{x} - \mathbf{x}_d$ is contracting under the input saturation and external disturbance.

Using state-dependent coefficient (SDC) formulation, the error dynamics can be derived in LPV form as follows:

$$\begin{aligned} \frac{d}{dt} \mathbf{e} &= \underbrace{\mathbf{f} - \mathbf{f}_d + (\mathbf{B} - \mathbf{B}_d)\mathbf{u}_d}_{=:\mathbb{A}(\mathbf{x}, \mathbf{x}_d)\mathbf{e}} + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \underbrace{\mathbf{B} \text{sat}(\mathbf{u}) - \mathbf{u}}_{=:\phi(\mathbf{u})} + \mathbf{d} \\ &= \mathbb{A}(\mathbf{x}, \mathbf{x}_d)\mathbf{e} + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d}, \end{aligned} \quad (10)$$

[1] H. Tsukamoto, S.-J. Chung, and J.-J. E. Slotine, "Contraction theory for nonlinear stability analysis and learning-based control: A tutorial overview," *Annual Reviews in Control*, 2021

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 \end{aligned} \tag{10}$$

Consider the following control law:

$$\mathbf{u} = \mathbf{u}_d - \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M}(\mathbf{x} - \mathbf{x}_d), \tag{11}$$

where $\mathbf{M}$ is a **contraction metric** which will be obtained.

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If the control law can be realized without input saturation, the error dynamics can be rewritten as follows:

$$\frac{d}{dt} \mathbf{e} = \left(\mathbb{A}(\mathbf{x}, \mathbf{x}_d) - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M} \right) \mathbf{e} + \mathbf{d}. \tag{12}$$

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Then, the steady-state tracking error is defined as follows:

$$\lim_{t \rightarrow \infty} \|\mathbf{e}(t)\| \leq \frac{\bar{d}}{\alpha} \sqrt{\chi}, \quad (\text{Theorem 2.4 in}^{[1]} \text{ and } \chi \text{ is cond. number of } \mathbf{M}) \quad (13)$$

[1] H. Tsukamoto, S.-J. Chung, and J.-J. E. Slotine, "Contraction theory for nonlinear stability analysis and learning-based control: A tutorial overview," *Annual Reviews in Control*, 2021

Revisit of Existing Approach (CV-STEM)

The LMI problem can be formulated to minimize the condition number χ of the contraction metric $\mathbf{M}$ as follows:

$$\begin{aligned} & \min_{\overline{\mathbf{W}}, \chi, \nu} \chi + \lambda \nu \\ \text{s.t. } & \begin{cases} -\frac{d}{dt} \overline{\mathbf{W}} + 2 \text{sym}(\mathbb{A} \overline{\mathbf{W}}) - 2\nu \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \preceq -2\alpha \overline{\mathbf{W}}, \\ \mathbf{I}_n \preceq \overline{\mathbf{W}} \preceq \chi \mathbf{I}_n, \end{cases} \end{aligned} \quad (14)$$

where $\mathbf{M} =: \overline{\mathbf{W}}^{-1}$, $\overline{\mathbf{W}} = \nu \mathbf{W}$ is normalized $\mathbf{W}$, and ν is decision variable for magnitude of the contraction metric, and λ is penalty parameter for ν .

- Solving the LMI problem online can be computationally expensive, depending on the system dimension.
- Input saturation cannot be directly incorporated.

Our Approach

We relax the contraction condition by projecting it onto the tracking error space.

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The contraction condition from Theorem 2.4 ^[1] is represented as follows:

$$\underbrace{\frac{d}{dt} \mathbf{M}}_{\approx (\mathbf{M} - \mathbf{M}_{\text{pre}}) / T_s} + 2 \text{sym}(\mathbf{M} \mathbf{A}) - 2 \mathbf{M} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^{\top} \mathbf{M} \preceq -2\alpha \mathbf{M}. \quad (15)$$

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We note that eventually the metric $\mathbf{M}$ is used as a feedback gain matrix as follows:

$$\mathbf{u} = \mathbf{u}_d - \mathbf{R}^{-1} \mathbf{B}^\top \mathbf{M} \mathbf{e}. \quad (16)$$

Therefore, the metric $\mathbf{M}$ enters the control input only through its action on the error $\mathbf{e}$.

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Thus, we can project the contraction condition onto $\mathbf{e}$, which results in the following concave condition:

$$\mathbf{e}^\top (\text{Eq. (15)}) \mathbf{e} \rightarrow \mathbf{y}^\top \left(-\mathbf{B} \mathbf{R}^{-1} \mathbf{B}^\top \right) \mathbf{y} + \mathbf{e}^\top \left(\frac{1}{T_s} + 2\alpha + 2\mathbb{A}^\top \right) \mathbf{y} - \frac{\mathbf{e}^\top \mathbf{M}_{\text{pre}} \mathbf{e}}{T_s} \leq 0, \quad (17)$$

where $\mathbf{y} := \mathbf{M} \mathbf{e}$ denotes a new decision variable which is a metric-weighted error vector.

[1] H. Tsukamoto, S.-J. Chung, and J.-J. E. Slotine, "Contraction theory for nonlinear stability analysis and learning-based control: A tutorial overview," *Annual Reviews in Control*, 2021

Obj Func.: Minimization of the condition number χ of $\mathbf{M}$

$$J = \|\mathbf{y}\| \|\mathbf{e}\| - \mathbf{y}^\top \mathbf{e} \quad (\text{promotes } \mathbf{M} \text{ to be isotropic — minimizing } \chi) \quad (18)$$

Cstr. 1 (Projected) Contraction Constraint:

$$\mathbf{y}^\top \left(-\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \right) \mathbf{y} + \mathbf{e}^\top \left(\frac{1}{T_s} + 2\alpha + 2\mathbb{A}^\top \right) \mathbf{y} - \frac{\mathbf{e}^\top \mathbf{M}_{\text{pre}} \mathbf{e}}{T_s} \leq 0. \quad (19)$$

Cstr. 2 Energy-Based Constraint:

$$\left(\mathbf{e}^\top \mathbf{y} - \mu \mathbf{e}^\top \mathbf{e} \right)^2 \leq (\epsilon + s) \|\mathbf{e}\|^2, \quad (\text{prevents } \mathbf{M} \text{ from being too small by obj. func.}) \quad (20)$$

where

- $\mu \in \mathbb{R}_{>0}$: desired energy level
- $\epsilon \in \mathbb{R}_{>0}$: allowable small energy variation
- $s \in \mathbb{R}_{\geq 0}$: slack variable (for feasibility)

Cstr. 3 Input Constraints: Any constraints can be added to the optimization problem.

$$c(\mathbf{y}) \leq 0 \quad (21)$$

Vector-Space Optimization Framework

Resulting Optimization Problem:

$$\begin{aligned}
 & \min_{\mathbf{y}, s} \|\mathbf{y}\| \|\mathbf{e}\| - \mathbf{y}^\top \mathbf{e} + \lambda s \\
 \text{s.t. } & \begin{cases} \mathbf{y}^\top (-\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top) \mathbf{y} + \mathbf{e}^\top \left(\frac{1}{T_s} + 2\alpha + 2\mathbf{A}^\top \right) \mathbf{y} - \frac{\mathbf{e}^\top \mathbf{M}_{\text{pre}} \mathbf{e}}{T_s} \leq 0, & \text{— (Projected) contraction constraint} \\ (\mathbf{e}^\top \mathbf{y} - \mu \mathbf{e}^\top \mathbf{e})^2 \leq (\epsilon + s) \|\mathbf{e}\|^2, & \text{— Energy-based constraint} \\ c(\mathbf{y}) \leq 0, & \text{— Input constraint} \end{cases} \quad (22)
 \end{aligned}$$

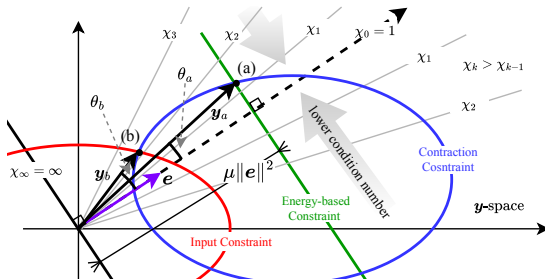
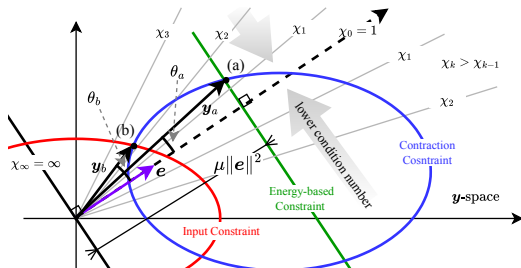


Figure: Example illustration of the proposed optimization problem.



Without Input Saturation: (a) is the optimal solution.

- Minimum condition number χ .
- Input energy constraint satisfied.

With Input Saturation: (b) is the optimal solution.

- Input saturation constraint satisfied.
- Minimum condition number under the input constraint.
- Slack variable s is non-zero, relaxing energy-based constraint.

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We employed the Lorenz system to validate the proposed method, as follows:

$$\underbrace{\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{=: \mathbf{x}} = \underbrace{\begin{pmatrix} \sigma(y - x) \\ x(\rho - z) - y \\ xy - \beta z \end{pmatrix}}_{=: \mathbf{f}(\mathbf{x})} + \text{sat} \left(\underbrace{\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}}_{=: \mathbf{u}} \right) + \mathbf{d}(t), \quad (23)$$

where $\sigma = 10$, $\rho = 28$, and $\beta = 8/3$.

The saturation function and external disturbance are defined as follows:

$$\text{sat}(\mathbf{u}) := \begin{cases} \mathbf{u}, & \text{if } \|\mathbf{u}\| \leq \bar{u}, \\ \bar{u} \frac{\mathbf{u}}{\|\mathbf{u}\|}, & \text{otherwise,} \end{cases}, \quad \text{and} \quad \mathbf{d}(t) := 1000(H(t - 0.2) - H(t - 0.21)) \cdot \mathbf{1}_{3 \times 1}, \quad (24)$$

where $\bar{u} = 160$ and $H(\cdot)$ is the Heaviside step function.

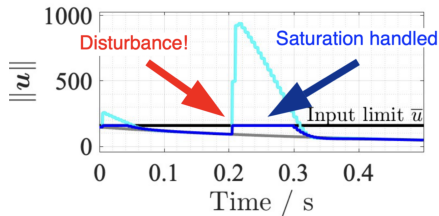
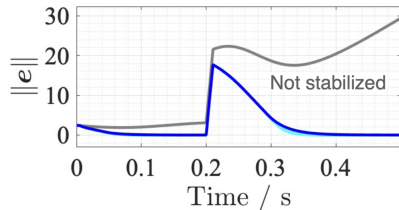
The desired trajectory is given by simulating the system with desired control law defined as follows:

$$\mathbf{u}_d(t) := -\mathbf{f}(\mathbf{x}_d(t)) - 2(\mathbf{x}_d(t) - \mathbf{r}), \quad \text{where } \mathbf{r} = \begin{pmatrix} -8.485 \\ 8.485 \\ 27 \end{pmatrix}. \quad (25)$$

Table: Controllers for comparative study.

(C ₁)	[—]	Proposed controller.
(C ₂)	[—]	CV-STEM ([1]).
(C ₃)	[—]	$\mathbf{u} = \mathbf{u}_d$.

- Using contraction theory, (C₁) and (C₂) stabilized the system under the external disturbance at 0.2 s.
- (C₂) produced excessive input, which is undesirable.
- In contrast, (C₁) successfully handled the input saturation.



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- Contraction condition is projected onto tracking error, resulting in vector-space optimization problem.
- Input saturation is directly incorporated into the optimization problem.

Future work

- Additional constraints, such as state constraints and CBF-motivated constraints, and input constraints will be considered.
- Since projected contraction condition is necessary but not sufficient, we will investigate the conditions under which the projected contraction condition is sufficient.

Thank you for your attention!

- [1] H. Tsukamoto and S.-J. Chung, "Neural contraction metrics for robust estimation and control: A convex optimization approach," *IEEE Control Systems Letters*, vol. 5, no. 1, pp. 211–216, 2021. [Online]. Available: <https://doi.org/10.1109/LCSYS.2020.3001646>